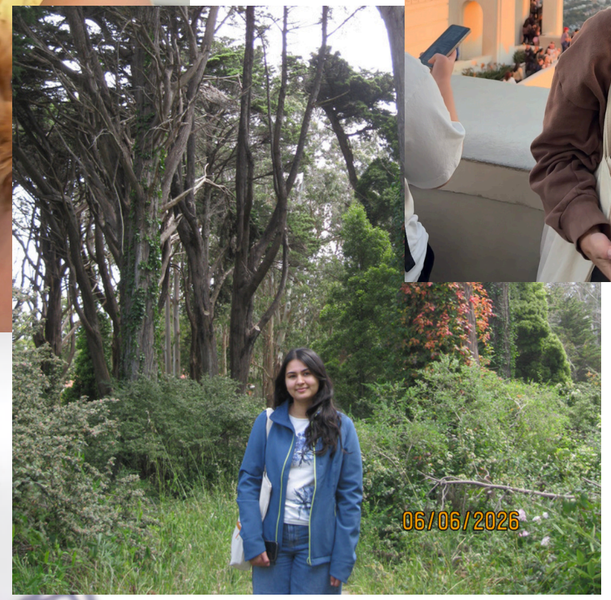
soundtrack

a wearable musical interface

ABOUT ME



Junior in ECE

2nd year in Makers

lovee music

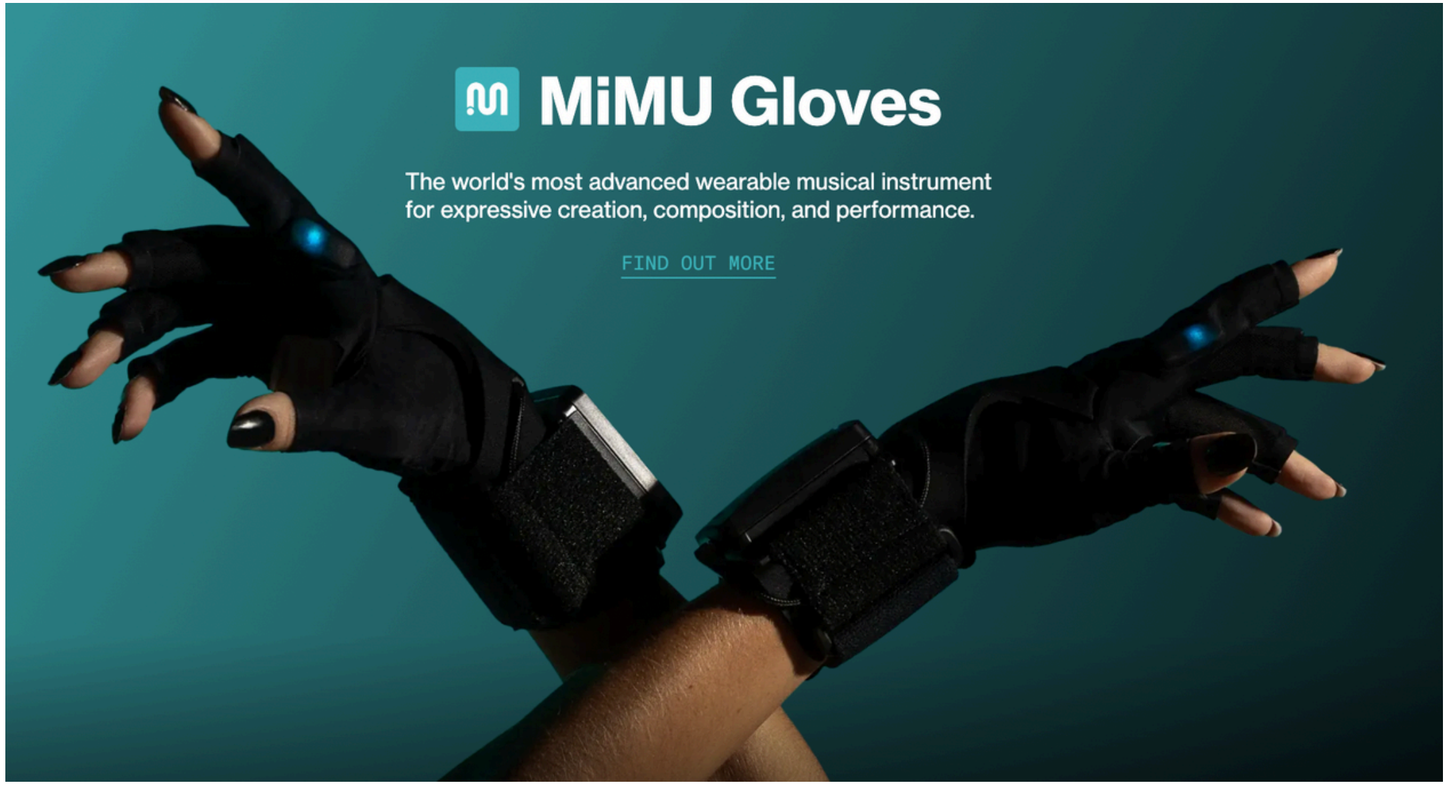
My fav artists rn: Noah Kahan, Tate McRae, Ethel Cain, Turnpike Troubadours

MOTIVATION

Traditional instruments require learning a physical interface.

So what if the instrument was just your hands ?

INSPIRATION



CONCEPT

01. MOVE

Bend • Tilt • Rotate • Gesture

02. INTERPRET

Sense + process movement

03. PLAY

Pitch • Volume • Timbre • Effects

SUBSYSTEMS

01. PRODUCT DESIGN

Wearable design
Sensor integration
Ergonomics & usability
Rapid prototyping

02. ELECTRICAL (ANALOG)

Analog circuits
PCB design
Power systems

03. ELECTRICAL (DIGITAL)

Sensor processing
Filtering & fusion
Motion interpretation
Audio/DSP exploration

04. SOFTWARE

Embedded firmware
Wireless systems
Audio & synthesis
System integration

TIMELINE:

Fall

SEPTEMBER

- Finalize team roles + architecture
- Select sensors/components
- Order parts ASAP

OCTOBER

- Test IMU + flex sensors
- Develop ESP32 firmware
- Establish wireless communication
- Begin sensor → sound mapping

NOVEMBER

- Build wearable single-glove prototype
- Sensing → audio pipeline
- Improve filtering + calibration
- Implement basic synthesis

DECEMBER

- Final testing + debugging
- Improve reliability + latency
- Demo single-glove

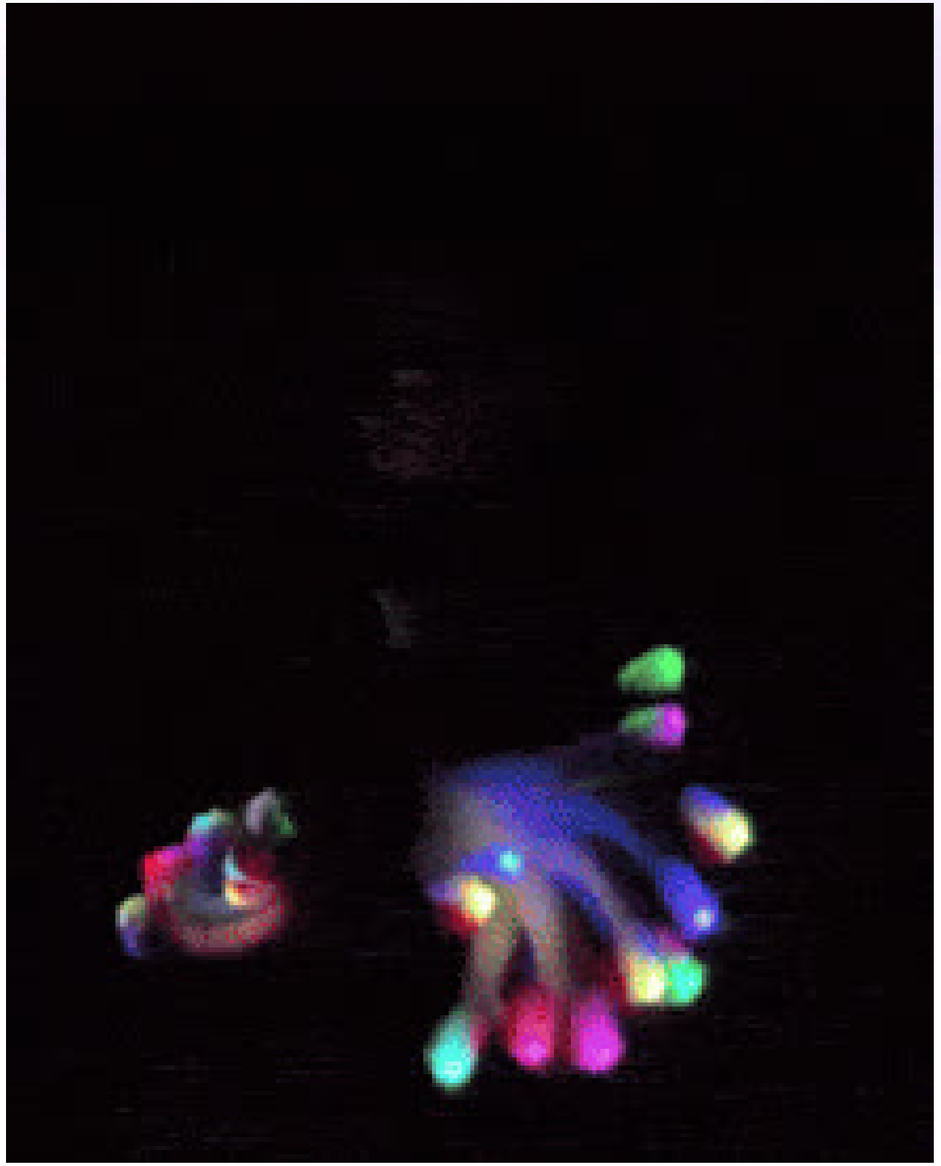
TIMELINE:

Spring

One Glove → Two Gloves → Expressive Instrument

WHY JOIN?

- build skills you'll actually use as an engineer
- partnership with Qualcomm using the Arduino Uno Q
- explore analog AND digital!
- exciting team bonding!
- i promise i will be a super fun PM 🙂



←you

THANK YOU!

Prital Jariwala